

DECODELABS

Industrial Training Kit — Batch 2026

Project 1

Data Cleaning & Preparation

Submission Report

Dataset: Dataset_for_Data_Analytics.xlsx

Records Audited: 1,200 | Fields Audited: 14

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1. Executive Summary

This report documents the data cleaning and preparation process performed on the dataset `Dataset_for_Data_Analytics.xlsx`, in fulfilment of Project 1 of the DecodeLabs Industrial Training Kit (Batch 2026). The objective of this milestone was to transform a raw dataset into a verified, production-ready source of truth by identifying missing values, eliminating duplicates, and correcting formatting issues.

The dataset was audited record-by-record and field-by-field using Python (pandas and openpyxl) for reproducibility. The audit found the dataset to be largely clean, with one significant data quality issue: 309 of 1,200 records (25.75%) were missing a `CouponCode` value. This was treated as a legitimate categorical gap (orders placed without a coupon) and resolved through category-level imputation rather than deletion, preserving all 1,200 records for analysis.

No duplicate Order IDs, no duplicate Tracking Numbers, no duplicate full records, and no incorrectly formatted dates were found. All numeric fields (`Quantity`, `UnitPrice`, `TotalPrice`) passed cross-validation. The dataset now meets the Project 2 Verification Gate threshold of a 0% error rate on unique identifiers and date formats.

2. Dataset Overview

The dataset contains 1,200 e-commerce order records across 14 fields, summarized below.

Column	Data Type	Description
OrderID	Text (ID)	Unique order identifier, format ORD#####
Date	Date	Date the order was placed
CustomerID	Text (ID)	Unique customer identifier, format C#####
Product	Categorical	Product purchased (7 categories: Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)
Quantity	Numeric	Units purchased per order (1–5)
UnitPrice	Numeric (Currency)	Price per unit
ShippingAddress	Text	Delivery address
PaymentMethod	Categorical	Payment method used (5 categories)
OrderStatus	Categorical	Current order status (5 categories)
TrackingNumber	Text (ID)	Shipment tracking code, format TRK#####
ItemsInCart	Numeric	Total items in the customer's cart at checkout
CouponCode	Categorical	Coupon applied, if any (3 active codes + blank for none)
ReferralSource	Categorical	Marketing channel that referred the order (5 categories)
TotalPrice	Numeric (Currency)	Order total; expected to equal $\text{Quantity} \times \text{UnitPrice}$

3. Methodology

Per the project's tooling guidance ('Choose Your Weapon: Scale Matters'), this dataset (1,200 rows) sits comfortably within Excel/Power Query scale; Python with pandas was used regardless to produce an auditable, reproducible script-based workflow rather than manual point-and-click edits. The following checks were run against every column and cross-checked against every row:

- Null/blank value scan across all 14 fields
- Exact duplicate row detection
- Regex pattern validation on ID fields (OrderID, CustomerID, TrackingNumber)
- Date validity and format consistency check (ISO 8601 standard: YYYY-MM-DD)
- Numeric range and sign check on Quantity and UnitPrice (negative/zero detection)
- Cross-field formula validation: TotalPrice vs. Quantity × UnitPrice
- Logical consistency check: ItemsInCart vs. Quantity
- Text consistency scan: leading/trailing whitespace and casing variants on all categorical and ID fields

4. Data Quality Audit Findings

Check	Method	Result	Status
Missing values (all fields)	isnull() scan, 1,200 rows × 14 fields	309 blanks in CouponCode only; 0 elsewhere	Action Taken
Duplicate OrderID	Exact + case/whitespace-normalized match	0 of 1,200	Pass
Duplicate TrackingNumber	Exact match	0 of 1,200	Pass
Duplicate full records	Row-level duplicate check	0 of 1,200	Pass
ID format validation	Regex: ORD#####, C#####, TRK#####	100% conformance	Pass
Date validity & format	Type + ISO 8601 consistency check	1,200 of 1,200 valid, range 2023-01-01 to 2025-06-30	Pass
Quantity / UnitPrice range	Negative or zero value check	0 invalid values	Pass
TotalPrice formula integrity	TotalPrice vs. Quantity × UnitPrice	0 mismatches across 1,200 rows	Pass
Text casing / whitespace	Whitespace trim + casing variant scan	0 inconsistencies across all categorical fields	Pass

5. Missing Value Treatment

5.1 Finding

CouponCode was blank in 309 of 1,200 records (25.75%). The three populated values (SAVE10, FREESHIP, WINTER15) are valid, well-formed promotional codes, so the blanks were assessed as orders placed without a coupon rather than as data-entry errors.

5.2 Decision & Justification

The training material's imputation guidance (Mean / Median / Mode) applies to numeric gaps. CouponCode is categorical, so a statistical average is not meaningful here. Listwise deletion was also rejected, per the explicit warning that it reduces statistical power and would have discarded over a quarter of the dataset.

Instead, all 309 blanks were imputed with the explicit category label "No Coupon". This preserves every record, keeps the field analysis-ready (e.g. coupon-usage rate, revenue by coupon type), and avoids an ambiguous null that downstream tools could misinterpret.

5.3 Result

CouponCode Value	Record Count	% of Dataset
FREESHIP	313	26.08%
No Coupon (imputed)	309	25.75%
WINTER15	292	24.33%
SAVE10	286	23.83%

6. Duplicate Removal

A full duplication audit was run at three levels: exact full-row duplicates, duplicate OrderID values (case- and whitespace-normalized), and duplicate TrackingNumber values. All three checks returned zero matches across all 1,200 records. No rows were removed, as none qualified as duplicates.

7. Format Standardization

7.1 Dates

All 1,200 Date values were validated as true date-type values (not text) and confirmed to fall within a plausible order window (2023-01-01 to 2025-06-30), with no future-dated or malformed entries. The column's cell format was standardized to ISO 8601 (YYYY-MM-DD) per the project's formatting standard.

7.2 Numbers

UnitPrice and TotalPrice were confirmed to carry consistent 2-decimal numeric precision with no currency symbols, thousands separators, or text-stored numbers. Quantity and ItemsInCart were confirmed as clean positive integers.

7.3 Text

All categorical fields (Product, PaymentMethod, OrderStatus, ReferralSource, CouponCode) and ID fields (OrderID, CustomerID, TrackingNumber) were scanned for inconsistent casing and leading/trailing whitespace ('Bangalore' vs 'BLR' vs 'bangalore'-style issues). None were found; no corrective action was required for these fields.

8. Verification Gate Results

Project 1 closes against the explicit threshold set for Project 2 readiness: zero duplicate IDs and zero incorrectly formatted dates.

Criterion	Required Threshold	Achieved Result
Duplicate Identifiers (OrderID)	0% error rate	0% — PASS
Duplicate Identifiers (TrackingNumber)	0% error rate	0% — PASS
Date Format Errors	0% error rate	0% — PASS
Missing Values Remaining	Fully resolved	0 remaining — PASS

9. Change Log

Per the project standard ('If it isn't documented, it didn't happen'), every action taken — and every check that required no action — is logged below.

Change ID	Description	Impact	Status
CR001	Standardized Date column to ISO 8601 (YYYY-MM-DD) format	Applied formatting consistency across 1,200 records	Resolved
CR002	Imputed missing CouponCode values with category label "No Coupon"	Preserved 309 records; field now fully analysis-ready	Resolved
CR003	Audited OrderID and TrackingNumber for duplicates	Confirmed 0 duplicate identifiers across 1,200 records	Verified — No Action Needed
CR004	Cross-validated TotalPrice against Quantity × UnitPrice	Confirmed 100% formula integrity, 0 mismatches	Verified — No Action Needed

Change ID	Description	Impact	Status
CR005	Audited categorical text fields for casing/whitespace issues	Confirmed 0 inconsistencies; no edits required	Verified — No Action Needed
CR006	Validated ID fields against expected regex patterns	Confirmed 100% conformance across all ID fields	Verified — No Action Needed

10. Conclusion

The dataset has been audited against every requirement in the Project 1 brief: missing values were identified and resolved, duplicates were checked and confirmed absent, and formatting was reviewed and standardized. The single substantive data quality issue — missing CouponCode values — was resolved through justified categorical imputation rather than deletion, in line with the project's guidance to handle gaps rather than discard data.

The cleaned file, Dataset_for_Data_Analytics_CLEANED.xlsx, is submitted alongside this report. The dataset meets the 0% error-rate threshold required to proceed to Project 2.

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